

## L30: Sensitivity Analysis

### Simulation Without a Black Box

# Outline

One-at-a-Time Analysis

Tornado Diagram

Morris Screening

$\mu$ -star vs sigma Plot

Input Uncertainty Propagation

Key Takeaways

# OAT: One-at-a-Time Sensitivity

## OAT Procedure

Fix all parameters at their nominal values. Vary one parameter at a time by  $\pm 20\%$  (or  $\pm 1\sigma$ ). Record the change in the response KPI.

**Example — clinic  $W_q$  at nominal  $\rho = 0.85$ :**

| Parameter  | -20% | Nominal | +20% |
|------------|------|---------|------|
| $\lambda$  | 11.3 | 18.4    | 31.7 |
| $\mu$      | 31.7 | 18.4    | 11.3 |
| $c$        | 18.4 | 18.4    | 8.1  |
| $CV_{svc}$ | 14.2 | 18.4    | 23.1 |

**Swing** =  $\max W_q - \min W_q$  for each parameter.

```
import numpy as np

params_nominal = {
    "lam": 14.2, "mu": 1/12.7,
    "c": 1, "cv": 0.61
}
deltas = [-0.20, 0.0, +0.20]
results = {}

for param, val in params_nominal.items():
    row = []
    for d in deltas:
        p = params_nominal.copy()
        p[param] = val * (1 + d)
        row.append(
            run_clinic(p, reps=30)
        )
    results[param] = row

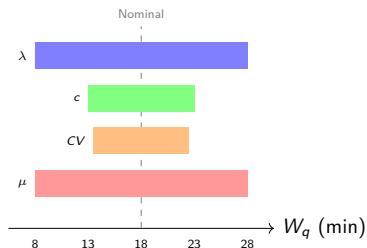
# Swing for each parameter
swings = {k: max(v)-min(v)
          for k,v in results.items()}
```

# Tornado Diagram: Ranking Parameters by Swing

## Construction:

1. Compute swing =  $W_q^{(+20\%)} - W_q^{(-20\%)}$  for each parameter
2. Sort parameters by |swing| descending
3. Draw horizontal bars centered at the nominal  $W_q$

| Parameter  | Swing (min) |
|------------|-------------|
| $\lambda$  | 20.4        |
| $\mu$      | 20.4        |
| $CV_{svc}$ | 8.9         |
| $c$        | 10.3        |



## OAT Limitation

OAT does not detect interactions. If  $\lambda$  and  $\mu$  move together (as in rush-hour), the true swing is not captured by varying each alone.

# Morris Screening: Elementary Effects

## Elementary Effect

For parameter  $x_i$  perturbed by  $\Delta$ :

$$EE_i = \frac{Y(\mathbf{x} + \Delta \mathbf{e}_i) - Y(\mathbf{x})}{\Delta}$$

Compute  $r$  elementary effects per parameter at different base points  $\mathbf{x}^{(j)}$ .

## Summary statistics:

$$\mu_i^* = \frac{1}{r} \sum_j |EE_i^{(j)}| \quad \sigma_i = \text{std}(EE_i^{(1)}, \dots, EE_i^{(r)})$$

- **High  $\mu_i^*$ :** parameter has large overall influence
- **High  $\sigma_i$ :** nonlinear behavior or strong interactions

**Advantage over OAT:** Morris explores the full parameter space ( $r \times (k + 1)$  model evaluations instead of  $2k + 1$ ).

```
import numpy as np

def morris_ee(model, x_nom, deltas,
              r=10, seed=42):
    rng = np.random.default_rng(seed)
    k = len(x_nom)
    ees = {i: [] for i in range(k)}

    for _ in range(r):
        # Random base point
        base = rng.uniform(0.2, 0.8, k)
        x0 = x_nom * base / 0.5
        y0 = model(x0)
        for i in range(k):
            x1 = x0.copy()
            x1[i] += deltas[i]
            y1 = model(x1)
            ees[i].append(
                (y1-y0)/deltas[i])

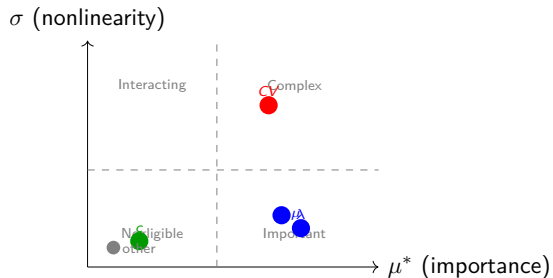
    mu_star = {i: np.mean(np.abs(v))
               for i,v in ees.items()}
    sigma = {i: np.std(v)
             for i,v in ees.items()}
    return mu_star, sigma
```

## $\mu^*$ vs. $\sigma$ Plot: Importance and Nonlinearity

### Four quadrants:

- Low  $\mu^*$ , low  $\sigma$ : **negligible** — safely fixed
- High  $\mu^*$ , low  $\sigma$ : **important, linear** — collect more data on this parameter
- High  $\mu^*$ , high  $\sigma$ : **important, nonlinear/interacting** — investigate further with full factorial
- Low  $\mu^*$ , high  $\sigma$ : **interacts but unimportant alone** — watch for interactions with high- $\mu^*$  parameters

**Rule:** Prioritize data collection for parameters in the high- $\mu^*$  region — they have the greatest influence on output uncertainty.



# Input Uncertainty Propagation

**Scenario:** We don't know  $\lambda$  exactly — only that it is approximately 14.2 with uncertainty. Suppose  $\lambda \sim \mathcal{N}(14.2, 1.5^2)$ .

## Propagation procedure:

1. Sample  $M$  parameter vectors  $(\lambda_m, \mu_m, \dots)$  from their marginal distributions
2. Run the simulation model for each parameter vector
3. Summarize the resulting output distribution

```
M = 200
lam_samples = rng.normal(14.2, 1.5, M)
mu_samples = rng.normal(1/12.7, 0.01, M)

Wq_out = [run_clinic(lam=l, mu=m)
           for l,m in
           zip(lam_samples, mu_samples)]

p5 = np.percentile(Wq_out, 5)
p95 = np.percentile(Wq_out, 95)
print(f"Wq: [{p5:.1f}, {p95:.1f}]")
```

## Interpretation

The 5th/95th percentile range of  $W_q$  quantifies *input uncertainty* — distinct from the statistical uncertainty addressed by CIs.

| Uncertainty source     | Reduced by                     |
|------------------------|--------------------------------|
| Statistical (sampling) | More replications              |
| Model form             | Better data / domain expertise |
| Input parameter        | Better data collection         |

**High  $\mu^*$  parameters drive input uncertainty.** Invest data-collection resources there first.

## Key Takeaways

1. **OAT tornado**: quick visual ranking; swing = max — min across the  $\pm 20\%$  range. Easy to implement, misses interactions.
2. **Morris  $\mu^*$** : average absolute elementary effect over the full parameter space;  $\sigma$  flags nonlinearity.
3.  **$\mu^*$  vs.  $\sigma$  plot**: four-quadrant diagnostic for importance and interaction structure.
4. **Input uncertainty propagation**: sample parameters from their marginal distributions; output 5th/95th percentile captures model risk.
5. **High  $\mu^* \Rightarrow$  collect more data**: sensitivity tells you *where* to spend the field-data budget.
6. **Sensitivity  $\neq$  decision importance**: domain knowledge can override statistics (a parameter may be “sensitive” but uncontrollable, or “insensitive” but ethically critical).

## Next Lecture

L31 — Ranking and selection: choosing the best system with guaranteed probability of correct selection.